

Question	Marking Guidance	Mark	Comments
04.1	1. <u>Osmosis</u> does not occur; 2. Chloroplast/organelle does not burst/lyse/shrivel/shrink;	2	1. Accept: osmosis would occur if water potentials were not the same. 1 and 2, Accept: correct reference to osmotic lysis for 2 marks . 2. Accept: chloroplast would burst/lyse/shrivel/shrink if water potentials were not the same. 2. Reject: ' <u>cell</u> bursts/shrivels' 2. Ignore: damage to chloroplasts on its own is not enough for a mark. 2. Reject: becomes turgid/flaccid.
04.2	1. To show light does not affect <u>DCPIP</u> ; 2. To show chloroplasts are required;	2	Ignore: comparison with other tubes.
04.3	1. Reduction of DCPIP by electrons; 2. (From) chlorophyll/light dependent reaction;	2	1. Accept: hydrogen/H for electrons but not protons/hydrogen ions/H ⁺ on their own. 2. Accept: from chloroplasts/photosystems/water.

04.4	Provides a standard / reference point OR Can compare different chemicals/weed-killers OR Can compare different concentrations of chemicals/weed-killers;	1	Accept: decolourises quicker than 100% or saves time waiting for complete decolourisation. Note: comparisons must be qualified. Accept: find the most effective weed-killer or the most effective concentration. Accept: answers relating to cost effectiveness.
04.5	1. Less/no ATP produced; 2. Less/no reduced NADP produced; 3. Less/no GP reduced/converted to TP;	2 max	2, Accept: less/no NADPH/NADPH₂/NADPH + H
